Problem set 3

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Problem 1

Solve the problem at the end of Chapter 8

Problem 2

Consider the model we went through in class, where the consumption Euler Equation is summarized by the socalled Fisher equation

$$\hat{i}_t = \hat{r}_t + E_t\pi_{t+1}$$

In class we showed that if the government sets and exogenous interest rate $\{i_t\}$ there is an infinite number of solutions. One solution to this problem was the one suggested by McCallum (1983) that the interest rate responds to inflation which resulted in a unique bounded solution.. Consider a slightly modified rule relative to what was in class

$$\hat{i}_t = \hat{r}_t + \phi_\pi\pi_t$$

which also results in a unique bounded solution.

(i) following the same steps as before, under what conditions does this result in a unique bounded solution? Compare this to the policy rule studied in class

$$\hat{i}_t = \phi_\pi\pi_t$$

Which one performs better and why? Which one do you think better describes Fed Policy

(ii) Now consider the following policy rule

$$i_t = \rho i_{t-1} + \phi_\pi\pi_t$$

which is called a inertial policy rule. People often advocate for this rule because it can generate the same inflation outcome but with lower volatility of interest rate.

iii) Is there a unique bounded solution in this case? (Hint: To solve the model, without having to use matrix algebra, you can forward the policy rule, take expectation of both sides, and subsitute into the Fisher equation and solve forward for i_t)

iv) Compare the volatility of inflation and interest rate accross the two policy rue. How doe they depend on ρ and ϕ_π

v) Some people suggest that the Fed should use market forecast of inflation in formulating its interest rate policy. Yet other people argue that this may lead to problems because if the Fed targets the expectations of the market, this could lead to indeterminacy. Evaluate this debate by considering the following policy rule. Can you provide any intuition? [Hint: It leads to indeterminacy!]

$$i_t = \phi_\pi E_t\pi_{t+1}$$